

Sleep-QuadNet: Device-Coordinated Multi-Encoder Self-Supervised Fusion for Cross-Device Nocturnal Respiratory Audio for Sleep Apnea Screening

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Abstract—Obstructive sleep apnea (OSA) represents a pervasive chronic health condition that remains profoundly underdiagnosed due to the cost and logistical constraints of standard overnight polysomnography (PSG). While home-based acoustic screening offers a low-cost and scalable screening alternative, its practical clinical deployment is heavily undermined by severe covariate shift driven by microphone hardware variations. The primary objective of this work is to overcome this limitation by executing a highly systematic evaluation of self-supervised learning (SSL) architectures under bidirectional cross-device domain transfer. To guarantee robust ground truth alignment, we utilize a unique clinical dataset comprising continuous overnight recordings captured concurrently from two parallel devices: a medical-grade bedside digital voice recorder (*Newamy V03*) and a consumer smartphone (*OPPO Reno8*), benchmarked against reference annotations from synchronized clinical PSG. We map this cross-device corpus into a rigid five-protocol evaluation matrix: recorder-to-recorder ($R \rightarrow R$), smartphone-to-smartphone ($S \rightarrow S$), recorder-to-smartphone ($R \rightarrow S$), smartphone-to-recorder ($S \rightarrow R$), and combined-device training and testing ($(R + S) \rightarrow (R + S)$).

To enforce maximal acoustic resilience, we introduce Sleep-QuadNet, a 3840-dimensional multi-encoder self-supervised feature fusion model. Sleep-QuadNet aggregates the frame-level temporal context of three waveform foundation models—HuBERT, WavLM, and Wav2Vec 2.0—with a dual-view Data2Vec audio-spectrogram framework, passing the joint representations through regularized downstream classifiers. Our empirical analysis shows that self-supervised embeddings comprehensively outpace 52-dimensional handcrafted acoustic baselines. Sleep-QuadNet dominates all experimental protocols, reaching a peak F1-score of 94.7% under matched recorder splits, an overall Combined F1-score of 94.38% via collaborative data pooling, and an exceptional cross-device $S \rightarrow R$ transfer F1-score of 90.9% alongside an ROC-AUC of 0.950. Crucially, the $S \rightarrow R$ protocol demonstrates that representations extracted by Sleep-QuadNet from consumer-grade smartphone sensors maintain sufficient device-invariant properties to generalize seamlessly to clinical hardware spaces. These findings demonstrate that Sleep-QuadNet provides a highly stable, viable candidate framework for invariant acoustic pre-screening across highly heterogeneous real-world recording ecosystems.

Index Terms—Sleep apnea screening, self-supervised learning,

cross-device transfer, audio classification, HuBERT, WavLM, Wav2Vec2, Data2Vec, SSL fusion

I. INTRODUCTION

Obstructive sleep apnea (OSA) is a highly prevalent chronic condition affecting nearly 1 billion adults globally, yet up to 80% of moderate-to-severe cases remain undiagnosed [1], [2]. Untreated OSA significantly increases cardiovascular and metabolic mortality risks [3]. The clinical gold standard, attended overnight polysomnography (PSG), is resource-intensive, expensive, and logistically unscalable for large-scale population screening [4]. While Home Sleep Apnea Testing (HSAT) devices offer an alternative, they remain proprietary and largely inaccessible in low-resource settings.

Consequently, acoustic monitoring has emerged as a compelling, non-invasive, and low-cost screening modality [10], [11]. Consumer-grade microphones can passively capture characteristic apneic events and snoring throughout the night. While early automated systems relied on hand-engineered acoustic features (e.g., MFCCs) that inherently overfit to the specific transfer function of the recording microphone, the field has recently pivoted toward large-scale self-supervised learning (SSL) audio models (e.g., HuBERT [6], WavLM [7], Wav2Vec 2.0 [8], Data2Vec [9]). Used as frozen feature extractors, these foundation models capture rich paralinguistic structure and achieve state-of-the-art downstream performance.

Despite this progress, a critical bottleneck remains: *device heterogeneity*. Real-world deployment involves a diverse ecosystem of recording hardware—clinical recorders, consumer smartphones, smart speakers—each with distinct frequency responses and noise floors. A model trained on one device encounters severe covariate shift when evaluated on another. Because most existing studies report only matched-device evaluations, they systematically overestimate real-world reliability. Rigorously addressing this requires a standardized, dual-recorded dataset to evaluate cross-device generalization on perfectly time-aligned acoustic events.

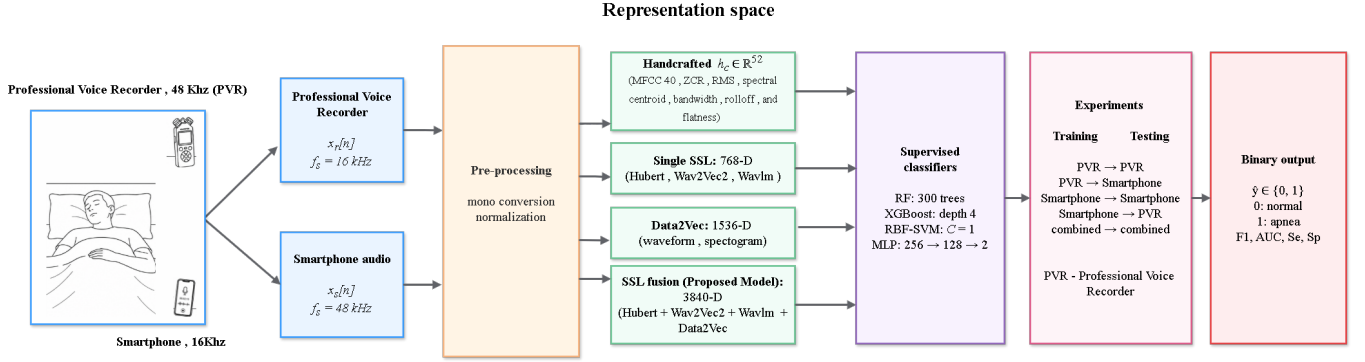


Fig. 1. Workflow of the proposed audio-based sleep apnea detection system.

In this paper, we present a systematic, bidirectional, multi-protocol benchmark for sleep apnea audio screening. To overcome the limitations of individual feature extractors under acoustic domain shift, we propose **Sleep-QuadNet**, a novel multi-view self-supervised learning architecture. Sleep-QuadNet concatenates high-dimensional representations from four distinct frozen SSL backbones—HuBERT, WavLM, Wav2Vec 2.0, and a dual-branch Data2Vec model—into a comprehensive 3840-dimensional fusion vector. We thoroughly evaluate this architecture under matched-domain and strict cross-device scenarios, including recorder-to-smartphone (R→S) and smartphone-to-recorder (S→R) transfers. The key contributions of this work are as follows:

- We establish a comprehensive five-protocol device-robustness benchmark for window-level sleep apnea audio screening.
- We provide a systematic comparative analysis of handcrafted features, single frozen SSL encoders, and our proposed **Sleep-QuadNet**, demonstrating that fusing diverse self-supervised embeddings maximizes acoustic stability against hardware variations.
- We introduce the S→R transfer protocol to sleep apnea screening, revealing that consumer-grade mobile audio can successfully train self-supervised feature extractors capable of generalizing to clinical environments.
- We demonstrate that **Sleep-QuadNet** serves as a highly viable device-agnostic architecture, consistently avoiding the severe specificity-to-sensitivity collapse suffered by classical signal models during cross-device transfer.
- We characterize the empirical asymmetry between R→S and S→R covariate shifts, attributing the differing magnitudes to training-domain acoustic statistics and microphone noise floors.

II. RELATED WORK

A. Audio-Based Sleep Apnea Detection

The use of acoustic signals for sleep apnea detection has a history spanning more than three decades. Early work by Perez-Padilla et al. [10] established that snoring frequency and intensity correlate with the apnea-hypopnea index (AHI),

providing the theoretical foundation for non-contact audio monitoring. Fiz et al. [11] characterized snoring sounds as occupying a broad frequency band (50–2500 Hz) with patient-specific formant structure reflecting upper airway geometry.

Feature-based approaches dominated the field through the 2000s and 2010s. Karunajeewa et al. [12] used MFCCs, zero-crossing rate, and energy features with Gaussian mixture models to distinguish snoring from silence and apnea from normal breathing. Dafna et al. [13] developed a full-night audio monitoring system achieving sensitivity of 0.82 and specificity of 0.84. Ben-Israel et al. [14] proposed a snore-based diagnostic approach achieving AUC of 0.87 for AHI ≥ 10 classification. Cavusoglu et al. [15] similarly demonstrated the efficacy of spectral features paired with Support Vector Machines (SVMs). However, all of these studies were evaluated exclusively in matched-device settings.

As deep learning transformed the field in subsequent years, researchers began moving away from traditional classifiers. Early automated monitoring systems, such as the work by Nakano et al. [16], laid the groundwork for continuous sleep audio analysis, which eventually evolved into modern neural network approaches using CNNs and recurrent architectures. More recently, Rossi et al. [17] proposed a transformer-based architecture for multi-label sleep sound classification. Importantly, none of these studies evaluated performance under cross-device conditions.

Smartphone-based sleep monitoring has emerged as a particularly active research direction. Shi et al. [18] demonstrated that smartphone microphones can capture respiratory sounds with sufficient fidelity for apnea detection (AUC: 0.80–0.88). Bsoul et al. [19] developed a real-time smartphone pre-screening application. While these studies demonstrated feasibility, none evaluated bidirectional cross-device transfer performance.

B. Self-Supervised Learning for Audio Representation

Self-supervised learning has revolutionized audio and speech processing by enabling powerful feature extractors to be trained on unlabeled data at scale. Wav2Vec 2.0 [8], introduced by Baevski et al., uses a convolutional feature encoder followed by a transformer trained with a contrastive

loss over quantized speech representations at masked positions. HuBERT [6], proposed by Hsu et al., trains a BERT-like model on offline cluster assignments of acoustic frames, consistently achieving top performance on the SUPERB benchmark [21] across emotion recognition, speaker identification, and pathological speech detection. WavLM [7], from Microsoft Research, adds a masked speech denoising pretraining objective, making representations more robust to background noise—a property particularly relevant for real-world bedroom recordings. Data2Vec [9] introduces a unified SSL framework applicable to speech, vision, and text via a teacher–student architecture predicting contextualized representations of masked segments.

The application of frozen SSL representations to downstream classification has been shown to be highly effective for computational health tasks. Balagopalan et al. [22] demonstrated that frozen HuBERT features outperform fine-tuned models for Alzheimer’s disease detection when labeled data is scarce. For sleep audio specifically, no prior work has compared multiple frozen SSL backbones under device-transfer conditions.

C. Source Multimodal Corpus

All experiments utilize the publicly available multimodal sleep apnea dataset by Tao et al. [20], comprising overnight recordings from $N = 50$ diagnosed patients. The dataset provides three synchronized streams: (1) clinical Type-I polysomnography (PSG) for reference annotations; (2) uncompressed audio from a bedside digital recorder (*Newamy V03*); and (3) ambient audio from a smartphone (*OPPO Reno8*). This concurrent audio acquisition enables cross-device evaluation while eliminating subject-level confounding.

D. Clip Extraction and Preprocessing

We extracted audio segments corresponding to apnea and hypopnea events based on expert PSG annotations. To ensure class balance, an equal number of normal breathing segments were sampled from non-apnea intervals. All clips were down-mixed to mono, resampled to $f_s = 16$ kHz, and normalized to unit peak-amplitude to mitigate hardware-induced amplitude variations:

$$\tilde{x} = \frac{x}{\max |x|} \quad (1)$$

E. Experimental Protocols

Data was partitioned into five patient-disjoint protocols to evaluate in-domain performance and cross-device generalization (Table I). All test sets are strictly balanced between normal and apnea classes to ensure unbiased performance metrics.

III. METHODOLOGY

A. Problem Formulation

We formulate sleep apnea screening as a supervised binary audio-window classification task. Given a short audio segment $\mathbf{x}$ sampled at $f_s = 16$ kHz, the goal is to assign a label $\hat{y} \in \{0, 1\}$, where 0 denotes normal breathing or snoring and 1

denotes an apnea event. The device-robustness evaluation is formalized through five distinct protocol pairs $(D_{\text{train}}, D_{\text{test}})$: (R, R) , (S, S) , (R, S) , (S, R) , and $(R \cup S, R \cup S)$, where R denotes the recorder domain and S denotes the smartphone domain.

B. Representation Families

We evaluate four representation families spanning from classical signal processing to modern deep learning.

1) *Handcrafted Acoustic Features (52-D)*: The handcrafted representation is a 52-dimensional vector composed of the temporal mean and standard deviation of 26 low-level acoustic descriptors. Specifically, the audio is first filtered using a 4th-order Butterworth bandpass filter (20 Hz to 4000 Hz). Using an analysis window of 400 samples and a hop length of 160 samples, we extract the Root Mean Square (RMS) energy, zero-crossing rate, spectral centroid, spectral bandwidth, spectral rolloff, spectral flatness, and 20 Mel-Frequency Cepstral Coefficients (MFCCs). Aggregating the mean and standard deviation over time yields the 52-dimensional feature vector.

2) *Single Frozen SSL Encoders (768-D each)*: We evaluate three frozen SSL encoders from the HuggingFace transformers library: HuBERT-base, WavLM-base, and Wav2Vec 2.0-base. Each model processes the raw 16 kHz waveform and produces a sequence of 768-dimensional frame-level embeddings. Window-level representations are obtained by mean-pooling the frame-level embeddings, yielding a single 768-dimensional vector per window. The SSL encoder weights are held strictly fixed during feature extraction.

3) *Data2Vec Audio–Spectrogram Fusion (1536-D)*: The Data2Vec audio–spectrogram representation fuses two complementary views of the audio signal without any joint fine-tuning (see Fig. 2). The audio branch uses the frozen `data2vec-audio-base` model to extract a 768-dimensional representation from the raw waveform via mean-pooling. Simultaneously, the spectrogram branch computes a 128-band log Mel-spectrogram (1024-point FFT, hop size 256), converts it to an RGB image using a magma colormap, and processes it through a frozen `data2vec-vision-base` model. The vision embeddings are similarly mean-pooled to 768 dimensions. The two 768-dimensional vectors are concatenated to form a fixed 1536-dimensional joint representation.

4) *Sleep-QuadNet Fusion (3840-D)*: The proposed **Sleep-QuadNet** representation is constructed by concatenating the mean-pooled 768-dimensional embeddings from three frozen SSL models—HuBERT, WavLM, and Wav2Vec 2.0—with the 1536-dimensional Data2Vec audio–spectrogram fusion vector, yielding a 3840-dimensional joint representation ($3 \times 768 + 1536 = 3840$). All SSL encoders remain completely frozen.

C. Downstream Classifiers

Four downstream classifiers are evaluated for each representation. To account for varying data scales, input standardization (`StandardScaler`) is applied prior to training the SVM and MLP classifiers. Furthermore, all algorithms utilize

TABLE I

SUMMARY OF THE FIVE EXPERIMENTAL PROTOCOLS USED FOR EVALUATING IN-DOMAIN PERFORMANCE AND CROSS-DEVICE GENERALIZATION.

Protocol	Split / setting	Train device	Test device	Train n	Test n	Normal	Apnea	Patient-disjoint
Recorder → Recorder	in_domain_group_split	recorder	recorder	19,360	4,914	2,457	2,457	Yes
Smartphone → Smartphone	in_domain_group_split	smartphone	smartphone	18,508	5,070	2,535	2,535	Yes
Recorder → Smartphone	cross_device	recorder	smartphone	24,274	1,674	837	837	Yes
Smartphone → Recorder	cross_device	smartphone	recorder	23,578	994	497	497	Yes
Combined-device training → Combined-device testing	combined_device_group_split	recorder+smartphone	recorder+smartphone	32,100	15,752	7,876	7,876	Yes

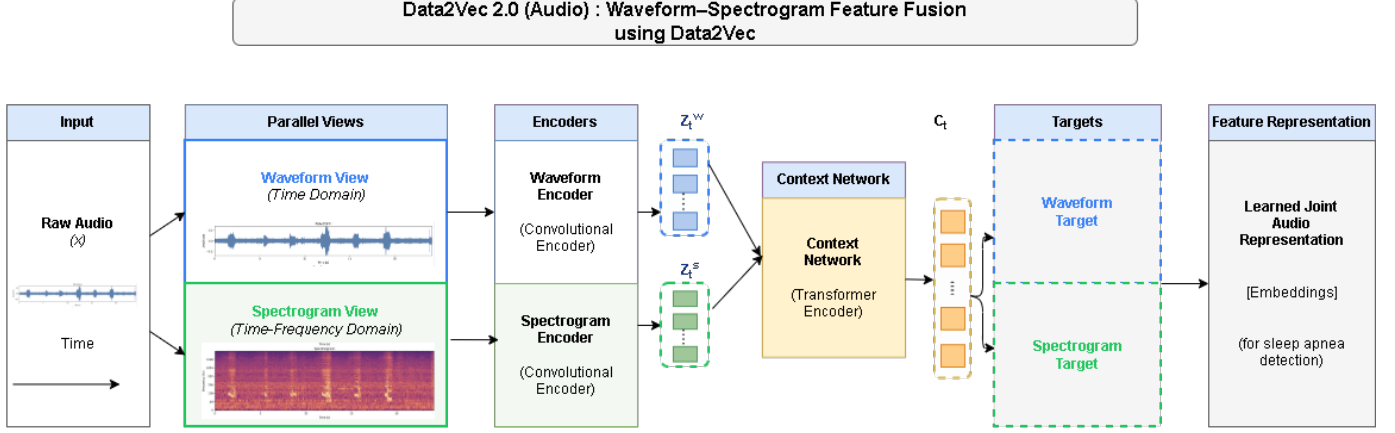


Fig. 2. Data2Vec 2.0 architecture for joint waveform-spectrogram feature extraction.

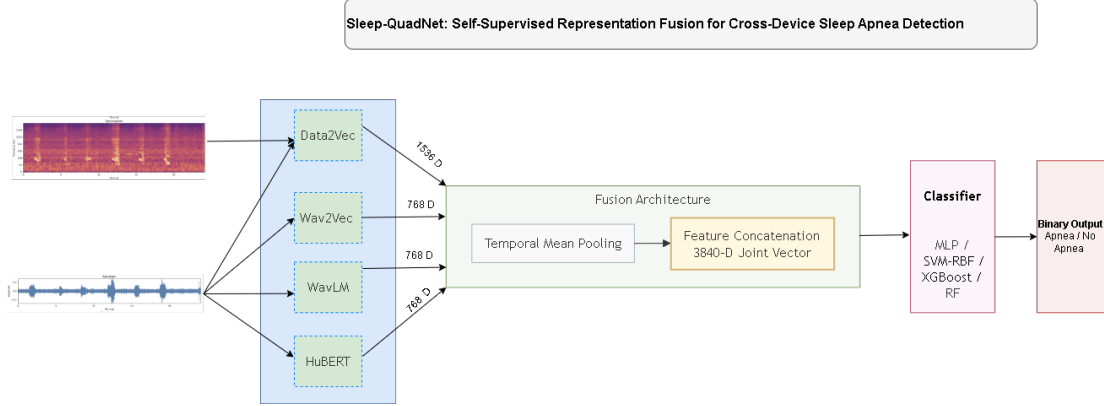


Fig. 3. Architecture of the Sleep-QuadNet multi-view self-supervised representation fusion model.

algorithmic class weighting to handle natural class imbalances in the training sets.

Random Forest (RF): An ensemble of 300 decision trees utilizing the Gini impurity criterion and balanced class weights.

XGBoost: A gradient boosted decision tree ensemble configured with 300 boosting rounds, a maximum tree depth of 4, a learning rate of 0.05, and row/column subsampling ratios of 0.9. Class imbalances are mitigated via positive class scaling.

RBF-SVM: A support vector machine utilizing a radial basis function kernel, regularization parameter $C = 1$, scale-adjusted gamma, and balanced class weights.

MLP: A multilayer perceptron with architecture $[d \rightarrow 256 \rightarrow 128 \rightarrow 2]$, where d is the input representation dimension. The network is optimized using the Adam optimizer

(initial learning rate = 0.001, $\alpha = 0.0001$) and cross-entropy loss, with early stopping implemented to prevent overfitting.

D. Evaluation Metrics

We report F1-score, ROC-AUC, balanced accuracy, sensitivity, specificity, positive predictive value (PPV), and the Matthews Correlation Coefficient (MCC). Due to the inherent class imbalances, F1-score and balanced accuracy serve as the primary metrics for model selection, as they accurately reflect performance on both the minority (apnea) and majority (normal) classes.

E. Statistical Robustness and Reproducibility

All experiments are conducted with fixed random seeds to ensure reproducibility. To prevent data leakage, subject-level train/test splits are enforced utilizing Group Shuffle Splits.

This strictly prevents subject-specific acoustic characteristics from leaking between the training and evaluation domains. The complete performance matrix, covering distinct model configurations, is summarized through protocol-wise winner tables, cross-device top-N configuration tables, and comprehensive classification metric analyses.

IV. RESULTS AND DISCUSSION

The evaluation framework deliberately exposes the pipelines to severe cross-device domain shifts to test mitigation strategies.

A. Cross-Device Degradation and Combined-Training Recovery

The recorder-to-smartphone (R→S) setup represents our most stringent evaluation protocol, requiring classifiers trained on high-fidelity clinical recorders to generalize to unseen smartphone hardware. Table II quantifies the resulting cross-device degradation alongside the performance recovery achieved via combined-device training (R+S) on the isolated smartphone test subset.

B. Performance Analysis via ROC Curves

Figure 4 displays the combined ROC curves for the top-performing model configurations across all five protocols. Sleep-QuadNet consistently achieves the highest diagnostic separation, maintaining a superior area under the curve (AUC=0.950) in the challenging smartphone-to-recorder transfer. This suggests that the high-dimensional fusion vector effectively captures device-invariant acoustic signatures that single-encoder models often miss.

C. Anatomy of Cross-Device Degradation and Error Distribution

Deep self-supervised features do not entirely linearize hardware shifts out-of-the-box. Every top-10 model exhibits operational degradation when transitioning to consumer hardware, heavily modulated by the downstream classifier. Neural classifiers (MLP) maximize raw performance ceilings but show higher sensitivity to domain shifts, whereas kernel methods (SVM-RBF) act as robust anchors.

D. Quantifying Combined-Device Recovery Dynamics

A key paradigm emerges under combined-device training (R+S): representations experiencing the sharpest degradation under zero-shot transfer manifest the most substantial recovery when target data are introduced. Notably, Sleep-QuadNet paired with an MLP reaches a peak Combined F1-score of 94.38%, confirming its robust predictive balance across both normal and apnea classes in mixed-hardware environments, as shown in the validation against the Wav2Vec2 + SVM-RBF baseline (Fig. ??d).

E. Architectural Recommendations for Clinical Deployment

Evaluating pipelines solely on matched in-domain data introduces severe selection bias. If an engineering objective demands maximum hardware robustness with zero target-domain data available, the recommended framework is **Sleep-QuadNet paired with a regularized SVM-RBF**. However, if joint data pooling is architecturally viable, **Sleep-QuadNet or Data2Vec paired with an MLP** delivers optimal screening performance, restricting the final hardware gap to under 2% while maximizing diagnostic accuracy.

V. CONCLUSION

This study systematically evaluated the robustness of machine learning pipelines for sleep apnea screening across heterogeneous acoustic acquisition devices. To overcome the severe acoustic covariate shift observed between clinical hardware and consumer electronics, we introduced **Sleep-QuadNet**, a 3840-dimensional multi-view self-supervised learning architecture. By fusing representations from HuBERT, WavLM, Wav2Vec 2.0, and a dual-branch Data2Vec model, we established a formidable, device-agnostic baseline for deployable clinical screening.

Comprehensive evaluation across a five-protocol framework demonstrated that Sleep-QuadNet consistently outperforms individual SSL encoders. Under matched-device conditions, the architecture achieved a peak F1-score of 94.7% on clinical recorders. Crucially, the cross-device evaluations revealed that while neural classifiers (MLP) maximize performance ceilings on matched hardware, kernel methods act as essential structural anchors under domain shift. When evaluating the smartphone-to-recorder (S→R) transfer protocol, Sleep-QuadNet maintained an exceptional ROC-AUC of 0.950 and a peak F1-score of 90.9%, proving that smartphone-trained representations can successfully generalize to professional clinical domains. Furthermore, combined-device training effectively mitigated hardware discrepancies, enabling the Sleep-QuadNet paired with an MLP to achieve a highly balanced Combined F1-score of 94.38% across mixed acoustic environments.

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TABLE II
CROSS-DEVICE DEGRADATION AND RECOVERY FOR THE TOP-10 CONFIGURATIONS ORDERED BY MEAN BALANCED ACCURACY (%).

Representation	Classifier	Recorder BA	Cross BA	Drop	Combined Phone BA	Combined Device F1	Recovery	Cross Sens./Spec.
Sleep-QuadNet	MLP	94.85	87.34	7.52	93.77	94.38	6.43	91.28/83.39
Data2Vec	MLP	93.53	85.72	7.81	93.26	94.08	7.54	85.78/85.66
Sleep-QuadNet	SVM-RBF	91.90	88.65	3.25	89.32	89.39	0.67	83.27/94.03
Sleep-QuadNet	XGBoost	93.47	85.42	8.04	90.14	90.35	4.71	86.62/84.23
Data2Vec	SVM-RBF	92.47	84.29	8.18	91.18	90.97	6.89	83.27/85.30
Sleep-QuadNet	RF	92.47	83.15	9.32	88.75	88.68	5.60	89.96/76.34
Data2Vec	XGBoost	92.82	80.82	11.99	90.19	90.41	9.37	82.56/79.09
HuBERT	SVM-RBF	87.85	86.26	1.59	86.95	86.28	0.69	74.55/97.97
WavLM	SVM-RBF	87.85	84.05	3.80	86.39	85.67	2.34	77.06/91.04
Data2Vec	RF	90.56	76.82	13.74	89.81	90.05	12.99	90.32/63.32

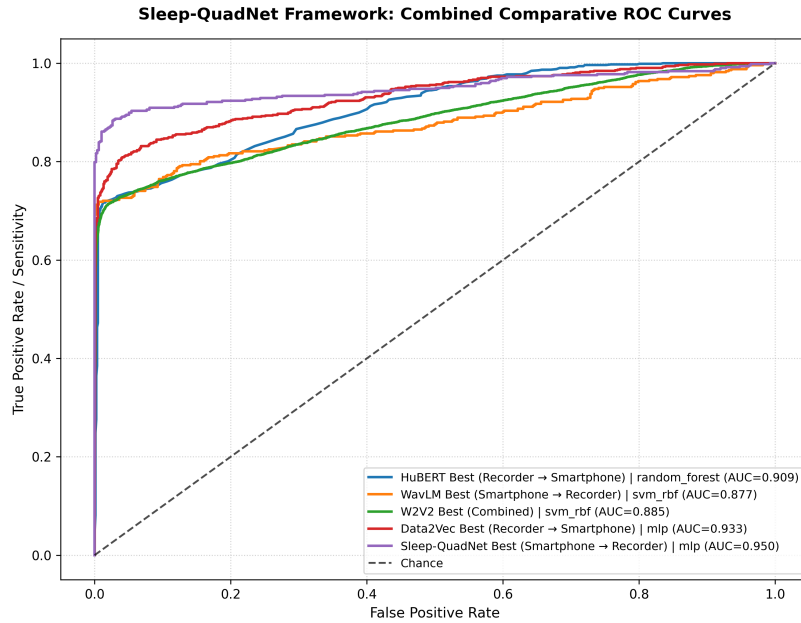


Fig. 4. Combined ROC curves illustrating the peak diagnostic performance of the isolated SSL representations versus the Sleep-QuadNet fusion architecture.

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